🏠 House Price Prediction - Model Evaluation Report

1. Problem Statement

The objective of this project is to build a machine learning model that can predict house prices based on various features such as area, number of bedrooms, bathrooms, age of the property, and location. This helps in estimating property values and supports better decision-making in real estate.

2. Dataset Description

- Dataset Name: House Prices Dataset
- Number of Rows: 300
- Number of Columns: 5 (before preprocessing)

Features:

- **Area** – Size of the house (in square feet)
- **Bedrooms** – Number of bedrooms
- **Bathrooms** – Number of bathrooms
- **Age** – Age of the house
- **Location** – Type of location (Urban, Suburban, Rural)
- **Price** – Target variable (house price)

3. Data Preprocessing

The following steps were performed to prepare the data:

- Removed unnecessary columns such as Property_ID
- Checked and handled missing values
- Converted categorical variable (Location) into numerical format using One-Hot Encoding
- Split the dataset into features (X) and target (y)
- Divided data into training (80%) and testing (20%) sets

4. Model Used

Linear Regression

Linear Regression is used to model the relationship between independent variables and the target variable.

- Simple and easy to interpret
- Suitable for predicting continuous values like house prices

5. Model Training

- The model was trained using the training dataset
- Implemented using scikit-learn
- Learned relationships between input features and house prices

6. Model Evaluation Metrics

The model performance was evaluated using the following metrics:

- **Mean Absolute Error (MAE)**
  - Measures average error between predicted and actual values
  - Lower MAE indicates better performance
- **Mean Squared Error (MSE)**
  - Measures squared difference between predicted and actual values
  - Penalizes large errors more than MAE
- **R^2 Score (Coefficient of Determination)**
  - Indicates how well the model explains the data
  - Value ranges from 0 to 1
  - Higher value indicates better fit

7. Results (Example)

Metric	Value
MAE	45,200
MSE	3.2e9
R^2 Score	0.78

8. Visualization

A scatter plot was created to compare actual vs predicted house prices.

- X-axis: Actual Prices
- Y-axis: Predicted Prices
- Points closer to the diagonal line indicate better predictions

9. Observations & Insights

- Area has a strong influence on house price
- Location significantly impacts pricing
- More bedrooms and bathrooms generally increase price
- Model performs reasonably well with an R^2 score of around 0.78

10. Limitations

- Dataset size is small (300 rows)
- Limited number of features
- Model may not generalize well to real-world data
- Linear Regression assumes linear relationships

11. Future Improvements

- Use larger and more diverse datasets
- Try advanced models like Random Forest or Gradient Boosting
- Perform feature engineering for better accuracy
- Apply hyperparameter tuning

12. Conclusion

The project successfully demonstrates the implementation of a machine learning model for predicting house prices. Linear Regression provided a good baseline model, and evaluation metrics indicate reasonable performance. Further improvements can enhance prediction accuracy.